Challenge

Today

Welcome to your next lesson: Challenge.

I'm preparing your personalized lesson based on what you've learned, what's next, and the best way to help you practice.

Keep it up, every step counts.

Ron, we’re doing a 6-round Object Detective challenge today 🎯

No teaching, just you applying what you already know.

Here’s Round 1 (of 6):

Imagine this tiny Python script:

```
Python
import random

coin_flipper = random.Random()
result = coin_flipper.choice(["heads", "tails"])
print(result)
```

In one plain sentence, describe what role coin_flipper plays in this program (don’t worry about perfect wording, just your idea).

Then we’ll go to Round 2.

coin flipper is an object based on the random module. it is

used to generate a heads or tails result using randoms built in methods

What do you think for Round 2? Pick the option that best matches each piece 👇

Round 2 (of 6) 🕵️

Using the same code:

```
python
import random

coin_flipper = random.Random()
result = coin_flipper.choice(["heads", "tails"])
print(result)
```

Which option correctly matches each piece with its role?

random
Random
coin_flipper
choice
result

A.
random is a module, Random is a class in that module, coin_flipper is an object (instance) of that class, choice is a method of that object, and result is the data returned by calling that method.

B.
random is an object, Random is a module, coin_flipper is a class, choice is the result, and result is a method.

C.
random is a class, Random is a module, coin_flipper is a function, choice is a variable, and result is an object.

D.
random is a function, Random is a method, coin_flipper is data, choice is a module, and result is a class.

Nice, Ron — that mapping is spot on 🎯

random really is the module you import.
Random is the class defined inside that module.
coin_flipper is an object (instance) created from that class.
choice is a method on that object.
result is just the data (a string, "heads" or "tails") returned by calling that method.
Round 3 (of 6) 👨‍🔬

Same code again:

Python
import random

coin_flipper = random.Random()
result = coin_flipper.choice(["heads", "tails"])
print(result)
In one short sentence:

What type of thing is ["heads", "tails"] here, and what is it used for in this program?

its type is a List and its used as parameters for the choice function to randomly pick one or theother

What do you think Round 4 is doing? Pick the best option below 👇

Round 4 (of 6) 👨‍🔬

Look at this tiny Python program:

python
import random

rng = random.Random()
number = rng.randint(1, 10)
print(number)
Which statement best describes what rng.randint(1, 10) is doing?

A.
Calling the module random and turning it into a number between 1 and 10.

B.
Calling the method randint on the object rng, which returns a random integer from 1 to 10 inclusive.

C.
Creating a new class named randint that stores numbers from 1 to 10.

D.
Changing the state of the random module so that it always prints the same number.

You nailed that one, Ron 🎯

rng is an object of type random.Random().
randint is a method on that object.
rng.randint(1, 10) returns a random integer between 1 and 10 (inclusive), which then gets stored in number.
Round 5 (of 6) 🤵

Now a quick “objects vs. data” check.

Look at this program:

Python
```
import random

rng = random.Random()
a = rng.random()
b = rng.random()
c = rng
```
In plain words, which names here refer to objects, and which refer to data values?

Just answer like: “Objects: ..., Data: ...”.

rng and c are objects
a,b, are data

Take your time with the question below 👇

Round 6 (of 6) 👮

You’re building a dice roller.

Which of these complete Python programs best follows the objects + methods pattern you’ve been using (module → object → method → returned data)?

A.
python
```
import random

roll = random.randint(1, 6)
print(roll)
```

B.
python
```
import random

die = random.Random()
roll = die.randint(1, 6)
print(roll)
```

C.
python
```
from random import randint

die = randint
roll = die(1, 6)
print(roll)
```

D.
python
```
import random

die = 6
roll = random(die)
print(roll)
```
Ron, that was excellent work 🎯

Your chosen program:

```
import random
```

```
die = random.Random()
roll = die.randint(1, 6)
print(roll)

follows the full chain:


random → module
Random() → creates an object
die → that object
die.randint(1, 6) → calling a method on that object
roll → the returned data (a random number 1–6)

That’s exactly the objects + methods pattern this unit wanted you to master ✅



You’ve completed the Challenge lesson for Thinking in objects 🎓


Strengths I see:


You clearly distinguish module / class / object / method / data.
You correctly track what’s an object vs a returned value in code.
You apply the pattern in fresh examples like coins and dice without hesitation.

Refinement to keep sharpening professionally:


Keep using precise wording (e.g., “method on an object that returns data”) when explaining your own code — that kind of clarity is gold in team environments and code reviews.

If you’re feeling good, go ahead and continue to the next lesson in Object-Oriented Programming — you’re ready for it 💪

How would you rate this lesson?


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